Design FMEA (qualitative version)

ED-Compact B-p smartphone cell - case exp/t6_r1_noceiling

VBF-T6R1NOCEILING-DFMEA-01

Design FMEA (qualitative version) - VBF-T6R1NOCEILING-DFMEA-01

Design under analysis: ED-Compact B-p (Chen2020 NMC811/graphite smartphone cell). Qualitative version, based on simulation signals (per deliverable-dfmea spec); severity/occurrence three-level qualitative ratings (H/M/L); RPN by simplified S×O matrix (qualitative caliber, annotated).

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
1	Lithium plating on negative electrode during 4C fast charge	Compaction (0.20 neg porosity, 85.2 µm, 4 µm particles) + baseline electrolyte transport cannot support 4C current	cell/r2_bp_4c.json:anode_potential_vmin = -0.2566 V -> plated (mechanical)	H (safety)	H (occurs at the advertised 4C)	Firmware charge derating (<= 2C recommendation, qualitative); electrolyte transport upgrade or anode coating - excluded in-boundary (ceiling_escalation OFF), recorded
2	Temperature limit exceed at 4C/45 °C	Heat load at 4C vs hA <= 0.133 W/K (A = 0.00531 m^2 locked, h <= 25)	cell/r2_bp_4c.json:T_max_K = 351.0 K > 323.15 K; h = 25 family datum 338.7 K still > limit	M	H (at 4C/45 °C)	Larger cooling envelope (pouch hA >= ~1.5 W/K) - excluded in-boundary, recorded; charge derating; ambient derating
3	Voltage plateau shortfall vs 4.1 V spec	NMC811 cathode OCP ceiling ~ 4.0 V (material property)	cell/r2_bp_energy.json:midpoint_voltage_v = 3.9214 V < 4.1 (best in-case 3.998 V)	L (spec compliance)	H (inherent to the system)	High-voltage cathode system (LNMO-class) - excluded in-boundary, recorded
4	Electrolyte oxidative decomposition	Upper voltage 4.2 V near NMC811 window edge	No HOMO/IE-EA data computed (real_compute = false - endorse skipped honestly); voltage window from parameter set only	M	L (<= 4.2 V window, qualitative)	Keep charge cut-off at 4.2 V (parameter-set value); DFT HOMO verification recommended in a real-compute run
5	Insufficient capacity vs nominal	Nominal (5.6 Ah) set from loading estimate, measured 5.3222 Ah (-5%)	cell/r2_bp_1c.json:capacity_ah	L	L	Datasheet rating reconciliation (report measured 5.32 Ah)
6	Excessive SEI growth / aging	Local current density in compacted negative	cell/r4_bp_aging.json:sei_thickness_nm_end = 417.74 nm <= 500 nm (pass); capacity trajectory climb-then-saturate artifact annotated	L	L	Pass at 1C/25 °C; 45 °C aging verification recommended (protocol available, not run for this candidate)

#	Failure mode	Failure cause	Simulation signal (detectability basis)	Severity	Occurrence	Design-side mitigation
7	Enclosure / tab / mechanical integrity failures	Outside simulation boundary	none	Not rated	Not rated	N/A (beyond pure-simulation boundary - complete FMEA incl. process/supplier failures N/A)

Conclusion

- Highest-risk items: #1 plating at 4C (S×O = H) and #2 thermal exceed at 4C/45 °C (M×H). Both are boundary-locked in this ablation (mitigation levers excluded: electrolyte formulation, coatings, cooling envelope redesign) - the honest in-boundary mitigation is charge derating, recorded in the datasheet fast-charge field.
- Mitigations implemented in the design itself: dead-layer thinning and compaction (ED objective), 4 µm negative particles (+0.022 V anode-potential margin, insufficient to clear 4C), SEI budget verified <= 500 nm.
- Complete FMEA (process/supplier failures): N/A (beyond pure simulation boundary).